

indicated in the list of formulae in page 3.

12 Ans: A

As the temperature is stable, the rate of increase in internal energy over the period is zero.

The filament, which would be hotter than the surroundings, would have lost heat, hence, the rate of heating of the filament would be negative.

By First Law of Thermodynamics, the rate of work done on the filament would be positive as work is done to drive unit charge through the filament.

Hence, A is the only possible answer.

13 Ans: D